

JAMESTOWN, ND, USA

WMO: 727535

Lat: 46.926N

Lon: 98.669W

Elev: 455

StdP: 95.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-25.3	-30.9	0.2	-27.5	-28.4	0.3	-25.1	14.9	-9.2	13.7	-10.1	4.2	310	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.8	22.3	30.0	21.1	28.4	20.1	23.9	29.3	22.7	27.9	21.5	26.6	5.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	18.0	27.1	21.0	16.6	26.1	19.6	15.2	24.6	74.3	29.4	69.3	28.0	64.8	26.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.6	10.4	DB	-31.1	35.4	3.1	1.9	-33.3	36.7	-35.1	37.8	-36.8	38.9	-39.0	40.2	(4)
(5)			WB	-31.3	26.1	3.0	1.3	-33.4	27.0	-35.2	27.8	-36.8	28.5	-39.0	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-12.2	-10.2	-2.9	5.4	12.6	18.5	21.2	19.8	15.0	7.0	-1.6	-9.0	(6)	
	DBStd	13.08	7.92	7.58	7.38	5.84	4.80	3.77	3.00	3.21	4.66	5.18	6.32	7.07	(7)	
	HDD10.0	2908	687	566	401	156	28	1	0	0	9	122	351	588	(8)	
	HDD18.3	4967	945	799	657	388	185	43	9	21	119	354	599	847	(9)	
	CDD10.0	1226	0	0	2	18	109	255	347	304	161	28	1	0	(10)	
	CDD18.3	243	0	0	0	1	8	47	98	68	21	1	0	0	(11)	
	CDH23.3	2517	0	0	1	14	126	476	943	678	254	25	0	0	(12)	
	CDH26.7	778	0	0	0	2	29	132	312	218	79	6	0	0	(13)	
(14) Wind	WSAvg	4.9	5.2	5.3	5.4	5.5	5.4	4.7	4.0	3.8	4.4	5.1	5.1	5.2	(14)	
(15) Precipitation	PrecAvg	452	15	13	22	39	57	84	75	54	42	26	13	11	(15)	
	PrecMax	767	55	55	88	78	158	194	281	146	134	110	62	31	(16)	
	PrecMin	243	0	0	0	0	6	24	10	7	0	1	0	1	(17)	
	PrecStd	107	13	15	20	22	41	42	55	36	32	25	14	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	7.8	18.8	25.4	29.9	32.6	34.2	33.8	32.3	27.0	17.5	7.6	(19)	
		MCWB	2.6	3.9	11.0	13.6	18.6	22.3	23.0	22.6	21.2	16.5	10.3	3.7	(20)	
	2%	DB	3.5	4.5	13.7	21.3	26.9	29.9	31.7	31.2	28.8	22.0	12.9	4.5	(21)	
		MCWB	1.2	2.2	8.4	11.7	16.6	20.6	23.1	21.6	18.9	14.0	7.5	1.7	(22)	
	5%	DB	1.5	2.5	9.6	18.1	24.1	28.0	29.9	29.0	26.3	18.2	9.8	2.3	(23)	
		MCWB	0.0	0.9	6.3	10.0	15.1	19.2	21.9	20.8	17.8	11.8	5.8	0.4	(24)	
	10%	DB	-0.7	0.9	6.3	15.2	21.7	26.2	28.2	27.3	23.7	15.3	6.8	0.3	(25)	
		MCWB	-1.7	-0.2	3.9	8.6	13.8	18.2	20.9	19.8	16.7	10.5	3.7	-1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	4.4	12.3	14.7	20.3	24.6	26.0	25.0	22.7	17.4	10.8	4.0	(27)	
		MCDB	5.3	7.1	16.4	22.8	26.9	29.9	30.9	30.3	29.6	24.1	16.3	7.2	(28)	
	2%	WB	1.4	2.5	8.9	12.3	18.2	22.5	24.3	23.3	20.8	14.8	8.2	1.9	(29)	
		MCDB	3.2	4.4	13.5	19.4	24.3	27.7	29.4	28.9	26.6	20.3	12.4	4.0	(30)	
	5%	WB	0.1	0.9	6.1	10.7	16.5	20.8	23.1	21.9	19.3	12.6	5.9	0.5	(31)	
		MCDB	1.5	2.2	9.5	17.3	21.7	26.2	28.1	27.0	24.2	17.6	9.6	2.1	(32)	
	10%	WB	-1.7	-0.4	3.9	9.2	14.9	19.4	21.9	20.7	17.8	10.5	3.7	-0.9	(33)	
		MCDB	-0.6	1.0	6.3	14.7	20.1	24.3	26.7	25.7	22.3	14.8	6.5	0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	9.8	11.8	12.2	11.3	12.0	12.7	12.8	10.9	9.7	8.5	(35)	
		MCDBR	11.0	10.5	14.8	17.6	15.9	14.2	14.3	15.3	16.9	16.8	14.6	11.4	(36)	
	5% WB	MCWBR	9.4	8.4	10.1	9.4	8.0	6.7	7.0	7.4	8.2	9.7	9.6	9.1	(37)	
		MCWBR	10.7	9.9	14.5	16.0	13.1	12.5	12.6	13.5	14.2	15.3	14.0	10.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.313	0.295	0.364	0.376	0.377	0.379	0.383	0.350	0.302	0.270	0.255	(40)	
		taud	2.248	2.254	2.461	2.287	2.331	2.377	2.380	2.355	2.425	2.537	2.509	2.446	(41)	
	Ebn at Noon	Ebn at Noon	773	847	937	889	889	887	879	861	862	866	834	814	(42)	
		Edh at Noon	80	98	93	123	122	117	115	114	97	74	61	58	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.68	2.80	3.93	4.95	5.66	6.27	6.52	5.47	4.13	2.60	1.63	1.24	(44)	
	RadStd	RadStd	0.15	0.29	0.32	0.39	0.42	0.42	0.32	0.42	0.27	0.33	0.21	0.11	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page